

CodeLit

A Windows Desktop and Web Application
for Secure Exam Lockdown and
Automated Programming Assessment

in Philippine Computing Education

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```
1 #include
2
3 int main() {
4     int n, i, sum = 0;
5     int current;
6
7     // Read N
8     scanf("%d", &n);
9
10
11     for(i = 0; i < n; i++) {
12         scanf("%d", &t);
13         sum += current;
14     }
15
16     printf("%d\n", sum);
17
18     return 0;
19 }
```

Run Sample Tests

Check Code

Test 1: Passed

Test 2: Passed

CONSOLE OUTPUT

```
$ gcc main.c -o solution
$ ./solution < input1.txt
150
Process finished with exit code 0
```

The Problem



No Desktop Lockdown

Cloud tools only restrict browsers — Alt+Tab, Task Manager, and OS-level shortcuts remain open.



Server-Side Bottlenecks

Simultaneous submissions to remote grading servers cause delays during time-limited exams.



Unreliable Connectivity

Many Philippine schools face unstable internet, making cloud-only platforms risky for live exams.



No Unified Platform

No single tool combines system-level security AND automated code assessment together.

Existing Solutions & Gaps

CodeChum

Strengths

- Automated code assessment
- Multi-language support (C, C++, Java...)
- Real-time performance monitoring
- SECUNA-certified, NPC-registered

Limitation

Browser-only, cannot enforce OS-level lockdown (Alt+Tab, Task Manager, Windows key).

Safe Exam Browser (SEB)

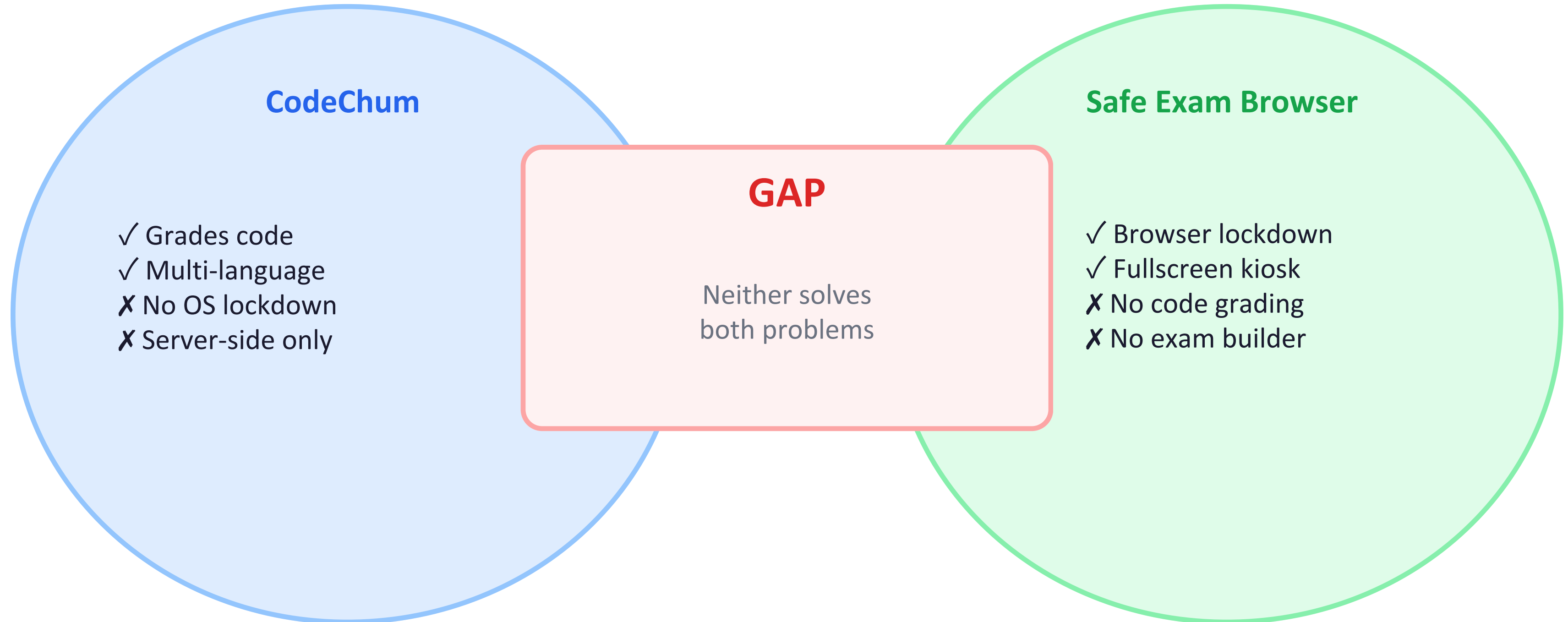
Strengths

- Enforces fullscreen, restricted browser
- Encrypted exam config files
- LMS integrations (Moodle, Canvas)
- Free and open-source

Limitation

No code evaluation, compilation, or programming assessment of any kind.

The Core Gap



No single platform currently provides secure OS-level lockdown + local code execution + automated assessment.

Proposed Solution

CodeLit

A Windows desktop application that combines system-wide exam lockdown with local automated programming assessment, supported by cloud-based exam management, synchronization, and monitoring.



System-Wide Lockdown

Intercepts Alt+Tab, Windows key, Task Manager, copy-paste, screenshots at OS level



Local Code Compiler

Compiles and executes C programs locally via GCC/MinGW — no server needed



Teacher Dashboard

Live monitoring, violation alerts, and session control from any browser



Multi-Type Exam Builder

Multiple Choice, Fill-in-the-Blank, and Coding questions in one exam



Cloud Sync & Storage

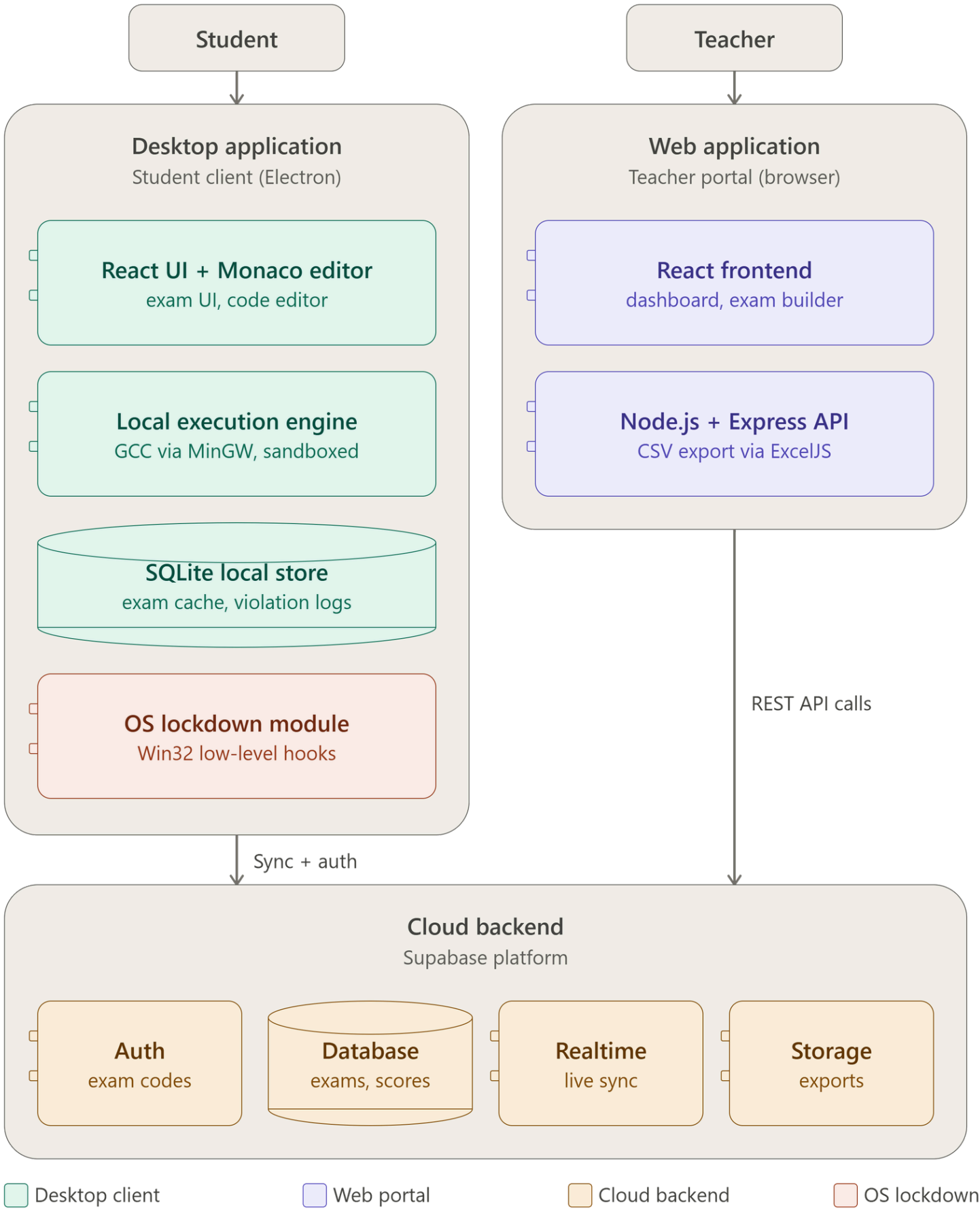
Centralized records, submissions, and violation logs stored securely



Export Reports

CSV result sheets and incident reports generated instantly after the exam

System Architecture



Two Entry Points

Student Side - Desktop App

- Monaco editor + local GCC/MinGW compiler (grades on-device)
- SQLite caches exam data locally
- OS Lockdown Module disables Alt+Tab, Start menu, copy-paste, Task Manager via Win32 hooks

Teacher Side - Web App

- React dashboard + exam builder
- Node.js/Express handles exam creation and CSV export

Cloud Backend

- Auth → single-use exam codes
- Database → exams, scores, violations
- Realtime → live teacher dashboard updates
- Storage → exported files

Key Design Principle

- Grading and lockdown happen locally, bandwidth doesn't affect exam performance
- The cloud's only job is keeping records accurate and the teacher's dashboard live

Lockdown & Security

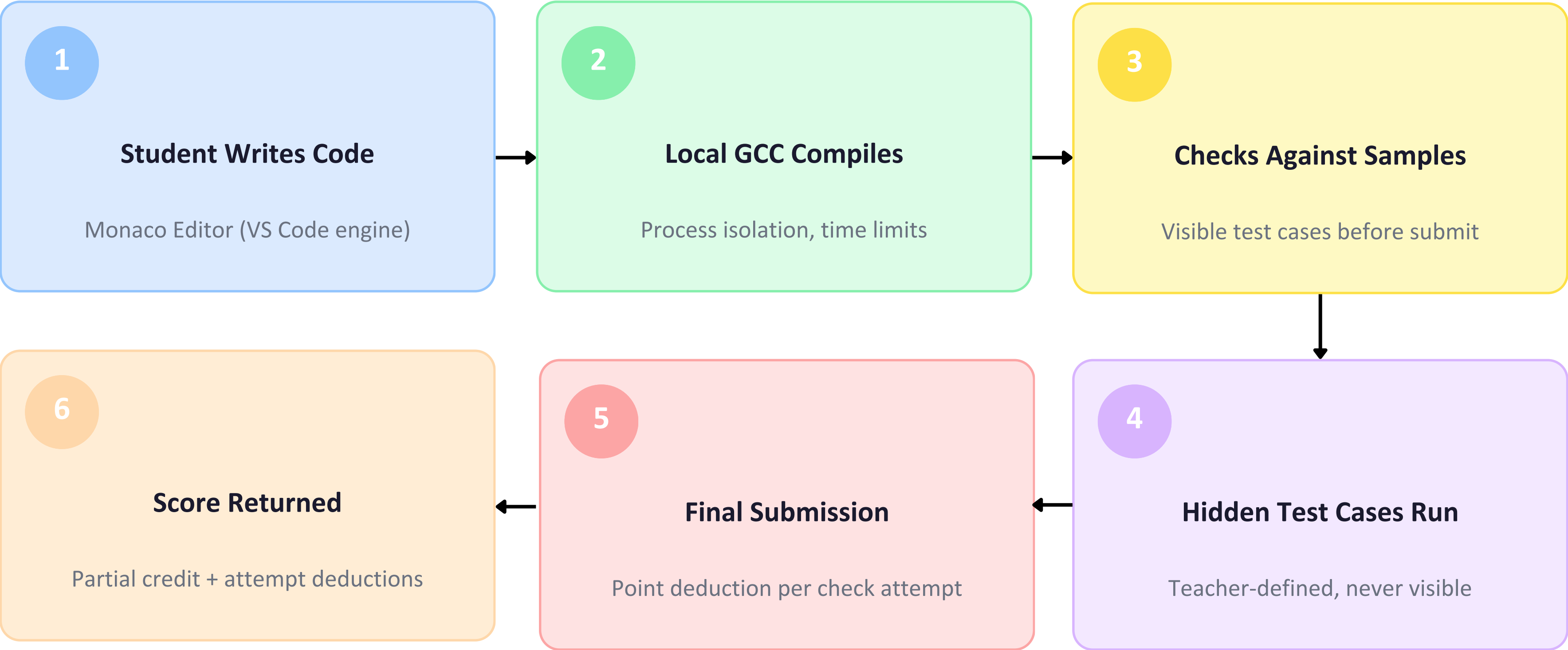
What's Intercepted & Blocked

- Alt+Tab and all app-switching shortcuts
- Windows key and Start menu
- Task Manager (Ctrl+Shift+Esc, Ctrl+Alt+Del)
- Copy-paste (Ctrl+C / Ctrl+V / Ctrl+X)
- Screenshot tools (PrintScreen, Win+Shift+S)
- Disallowed applications (process monitoring)
- System audio (muted on launch)

How It Works

- 1 Teacher starts exam → lockdown activates simultaneously on all student machines
- 2 SetWindowsHookEx (WH_KEYBOARD_LL) intercepts OS-level keyboard input in Ring 3 userspace
- 3 Fullscreen always-on-top kiosk window captures focus
- 4 Active process monitoring flags disallowed apps instantly
- 5 Bypass attempts silently logged with timestamps on teacher dashboard
- 6 Lockdown released only when teacher explicitly ends the session

Assessment Engine



How CodeLit Fills the Gap

Gap	CodeChum	SEB	CodeLit
OS-level Desktop Lockdown	✗	Partial	✓
Automated Code Grading	✓	✗	✓
Local Compilation (No Server)	✗	✗	✓
Coding-Specific Exam Builder	✓	✗	✓
Violation Logging & Reports	Partial	✗	✓
Works with Unstable Internet	✗	Partial	✓
Unified Exam Workflow	✗	✗	✓

CodeLit

Secure. Local. Integrated.

OS-Level Security

Local Code Execution

Cloud Management

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